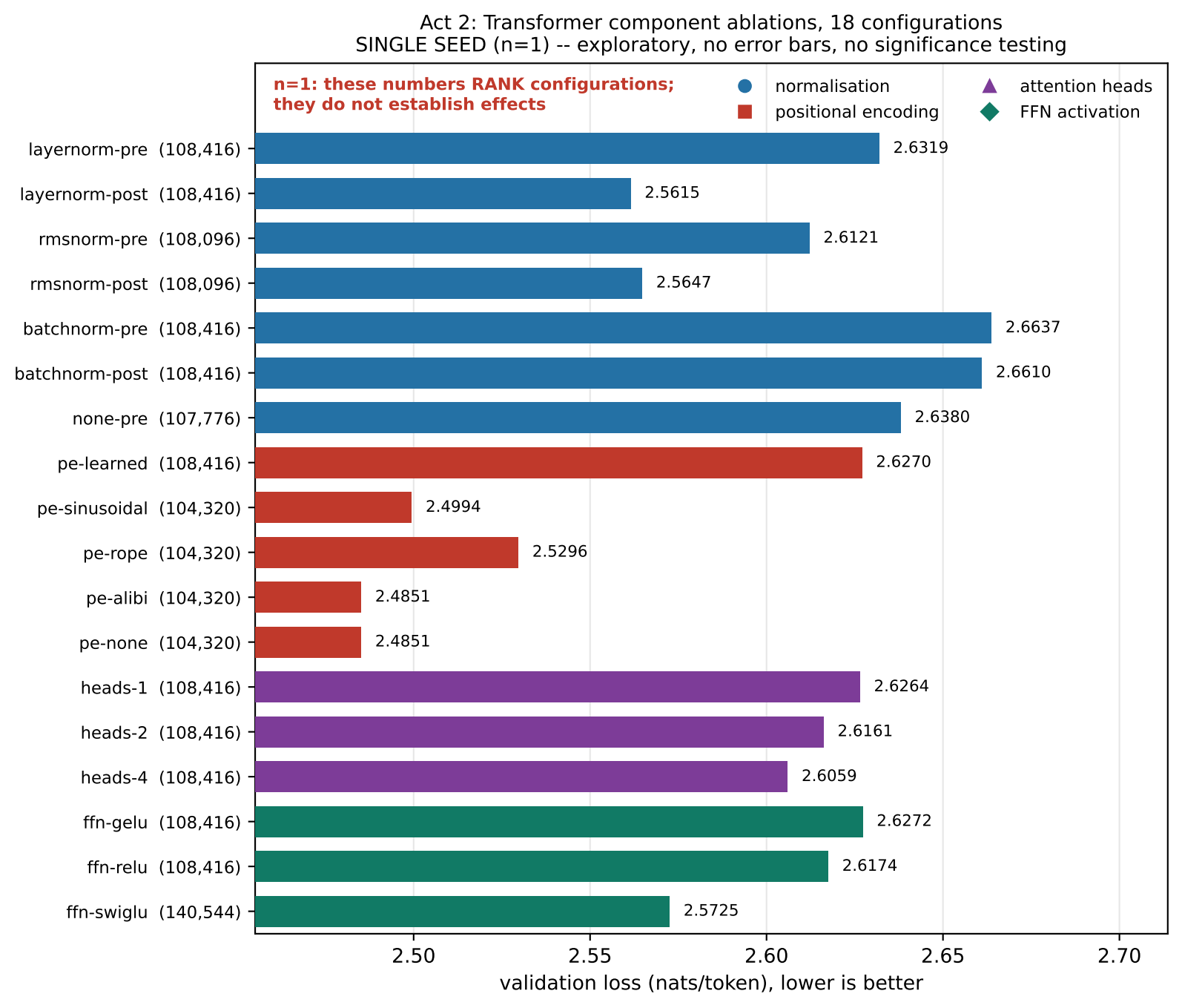
Act 2: Transformer component ablations, 18 configurations
SINGLE SEED (n=1) -- exploratory, no error bars, no significance testing
n=1: these numbers RANK configurations;
they do not establish effects
normalisation
positional encoding
attention heads
FFN activation
layernorm-pre (108,416)
layernorm-post (108,416)
rmsnorm-pre (108,096)
rmsnorm-post (108,096)
batchnorm-pre (108,416)
batchnorm-post (108,416)
none-pre (107,776)
pe-learned (108,416)
pe-sinusoidal (104,320)
pe-rope (104,320)
pe-alibi (104,320)
pe-none (104,320)
heads-1 (108,416)
heads-2 (108,416)
heads-4 (108,416)
ffn-gelu (108,416)
ffn-relu (108,416)
ffn-swiglu (140,544)
2.6319
2.5615
2.6121
2.5647
2.6637
2.6610
2.6380
2.6270
2.4994
2.5296
2.4851
2.4851
2.6264
2.6161
2.6059
2.6272
2.6174
2.5725
2.50
2.55
2.60
2.65
2.70
validation loss (nats/token), lower is better